HYDROCORTISONE

Adrenal crisis: give early

Sick-day rules

CHEMICAL NAME 11β,17,21-trihydroxypregn-4-ene-3,20-dione (cortisol); sodium succinate for injection
CLASS Corticosteroid (glucocorticoid with mineralocorticoid activity)
TRADE NAMES Solu-Cortef (IV) · Cortef · Efcortesol · Hydrocortone

THERAPEUTIC USES / INDICATIONS

- Adrenal insufficiency (Addison's, hypopituitarism, congenital adrenal hyperplasia); replacement.
- Adrenal crisis and steroid cover for surgery/illness in steroid-dependent patients.
- Acute severe asthma, anaphylaxis (adjunct), angioedema.
- Severe ulcerative colitis (IV); septic shock refractory to vasopressors; topical eczema; rectal foam.

ADMINISTRATION

- **IV** bolus over ≥1–10 min (or infusion); **IM**.
- **PO** replacement in divided doses (largest in the morning).
- **Topical** 0.5–1% cream; **rectal** foam/enema.

DOSAGES

- **Adrenal crisis:** 100 mg IV/IM immediately, then 200 mg/24 h (infusion or 50 mg every 6 h).
- **Replacement:** 15–25 mg/day PO in 2–3 doses (e.g. 10 mg on waking, 5 mg midday, 5 mg afternoon).
- **Sick-day rule:** double oral dose during fever/illness; IM if vomiting.
- **Acute asthma:** 100 mg IV every 6 h (if unable to take prednisolone).
- **Septic shock:** 200 mg/day.

PHARMACOKINETICS

- **Onset:** IV 1 h. **Duration:** 8–12 h (short-acting).
- Metabolised in liver (CYP3A4).
- Excreted in urine. **t½:** 1.5–2 h (biological 8–12 h).

PHARMACODYNAMICS

- Binds **glucocorticoid receptors** → anti-inflammatory, immunosuppressive, ↑ gluconeogenesis, ↑ vascular responsiveness to catecholamines.
- Significant **mineralocorticoid** effect → Na^+ and water retention, K^+ loss.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Hyperglycaemia, fluid retention.
- Mood change, insomnia.
- Increased appetite, dyspepsia.

ADVERSE EFFECTS (SERIOUS)

- Long-term: Cushing's syndrome, osteoporosis, adrenal suppression, diabetes, hypertension, cataract, infection risk, poor wound healing.
- Hypokalaemia.
- Psychosis.
- Peptic ulceration (with NSAIDs).

CONTRAINDICATIONS & CAUTIONS

- Systemic infection without antimicrobial cover (except life-threatening indications).
- Live vaccines with immunosuppressive doses.
- No absolute contraindication in adrenal crisis.

KEY INTERACTIONS

- **Enzyme inducers (rifampicin, phenytoin, carbamazepine):** ↓ effect (double replacement dose).
- **NSAIDs:** ↑ GI bleeding.
- **Diuretics, β2 agonists:** hypokalaemia.
- **Antidiabetics:** ↑ glucose.

NURSING MANAGEMENT

- Adrenal crisis: do not delay hydrocortisone for tests; IV fluids and glucose.
- Monitor BP, glucose, K^+, fluid balance.
- Make sure steroid-dependent patients never miss doses (NBM: switch to IV/IM).
- Long-term: weight, BP, glucose, bone health, infection.

PATIENT EDUCATION

- Carry a steroid emergency card and an emergency injection kit.
- Double your dose when ill with fever; inject if vomiting and seek help.
- Never stop suddenly.
- Take the largest dose on waking.

KEY POINTS

- Cortisol — glucocorticoid + mineralocorticoid.
- Replacement 15–25 mg/day; sick-day rules.
- Adrenal crisis: 100 mg IV/IM immediately.
- Never stop abruptly.

SOURCES CHECKED **[638]** Hydrocortisone 100 mg Powder for Solution for Injection or Infusion (UK SmPC) **[447]** BTS/SIGN asthma guideline (SIGN 158) **[381]** Evans et al., Intensive Care Med 2021

Full references p. 432–457